

How to treat autism in childrens?

Evidence searched on 2026-10-09 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line The retrieved evidence supports combining **FDA-approved antipsychotics (risperidone, aripiprazole) for irritability** with **behavioral therapies** as the primary treatment approach for autism in children. No evidence was found for medications targeting core autism symptoms (e.g., social communication, restricted interests) or non-pharmacological interventions beyond behavioral therapy.

Most important recent finding *No direct recent clinical trial or study results were retrieved.* The most recent cited source (FDA approval context) dates to **2023** (NICHD), confirming risperidone and aripiprazole's approval for **irritability in ASD children aged 5–17** (NICHD, 2023)

[\[https://www.nichd.nih.gov/health/topics/autism/conditioninfo/treatments/medication-treatment\]](https://www.nichd.nih.gov/health/topics/autism/conditioninfo/treatments/medication-treatment).

Immediate context Behavioral therapies are the **only established non-pharmacological approach** for autism in children, while antipsychotics are **FDA-approved only for irritability**, not core symptoms. No evidence was retrieved for other treatments (e.g., SSRIs, stimulants, or emerging therapies).

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly ..."	DailyMed

Latest findings

- **Advance publication (issue dated 2026 Dec): Case Reports** – Dual-target deep brain stimulation in a 12-month case study showed sustained reductions in core autism symptoms (ATEC, RBS-R, AuBC, CARS-2 scores improved by 65.9–96.4%) with no adverse events [1].
- **Advance publication (issue dated 2026 Dec): Systematic Review** – Extended reality (XR) interventions, particularly virtual reality (VR), demonstrated growing use for autism support, with hybrid systems and training/educational modalities most common [2].
- **Advance publication (issue dated 2026 Dec): Observational Study** – Sleep spindle characteristics (reduced amplitude/number, prolonged duration) were significantly associated with autism spectrum disorder [3].
- **2025 Jun: Randomized Controlled Trial** – Probiotic supplementation in ASD children increased microbiome diversity (Chao 1, Fisher's alpha, Shannon indices) without significant β -diversity changes [4].

Current evidence

Approach	Evidence level	What the evidence shows	Reference
Behavioral interventions (RCTs)	Evidence-supported but limited	Significant improvements in social-emotional and behavioral outcomes (Hedges' $g = 0.58$) for early childhood ASD.	Sandbank et al., <i>BMJ</i> , 2023 [5]
Pharmacological treatments	Emerging	Aripiprazole, atomoxetine, and bumetanide showed small-to-moderate effects on core ASD symptoms vs. placebo.	Siafis et al., <i>Molecular Autism</i> , 2022 [6]
Gluten-free/casein-free diet (GFCF)	Insufficient or negative	Mixed evidence: GFCF diets <i>may</i> reduce stereotypical behaviors and improve cognition, but most studies were small/blinded.	Quan et al., <i>Nutrition Reviews</i> , 2022 [7]
Music therapy	Emerging	Short- to medium-term benefits (3 days–8 months) for autistic children (ages 2–12) in individual/group settings.	Geretsegger et al., <i>Cochrane Database</i> , 2022 [8]
Extended reality (XR) interventions	Emerging	VR/AR systems (e.g., hybrid training/therapy) show promise but lack large-scale RCT validation.	Hao et al., <i>International Journal of Medical Informatics</i> , 2026 [2]
Emotion regulation interventions	Emerging	Non-pharmacological interventions improved emotion regulation/dysregulation in ASD youth.	Restoy et al., <i>Clinical Psychology Review</i> , 2024 [9]
Sensory integration therapy	Emerging	No significant effects on primary outcomes in a randomized trial ($n=138$), but carers reported functional improvements.	Randell et al., <i>Health Technology Assessment</i> , 2022 [10]

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Doses appear only where an FDA label or source gives them for the use shown; each dose entry is a short summary of the label's dosing section for this condition, with every number checked against the label text (full text in the DailyMed link); individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)." [associated symptom or comorbidity, not the condition itself]	Route: Oral. Start with 0.25 mg per day for children weighing less than 20 kg, or 0.5 mg per day for those weighing at least 20 kg. After no less than four days, increase to 0.5 mg per day for under-20 kg patients or 1.0 mg per day for those greater than or equal to 20 kg, and maintain this dose for at least 14 days. If insufficient response occurs, raise the dose by 0.25 mg per day for under-20 kg patients or 0.5 mg per day for those greater than or equal to 20 kg, at intervals of 2 weeks or greater. The usual dose range is 0.5 mg to 3 mg per day, and this applies to children weighing no less than 15 kg.	FDA-approved for the use shown; label effective 2026-06-25; original approval date not retrieved	FDA label: Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years). DailyMed label

Other drugs: used for symptoms (not established treatments of this condition)

Drugs that a retrieved review, guideline or web page describes as used for the condition, its symptoms or co-occurring conditions, for which no FDA label for this use was retrieved. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Key evidence
Aripiprazole	prescribed for irritability (population: children between 6 and 17 years old)	"Aripiprazole can be prescribed for children between 6 and 17 years old." Autism Spectrum Disorder (ASD) Treatment
Melatonin	addressing sleep problems in ASD [associated symptom or comorbidity, not the condition itself]	"Melatonin is commonly used to address sleep problems in ASD and appears to be effective." Pharmacological treatments in autism spectrum disorder: a narrative review / Journal of Psychopathology
Methylphenidate	treating ADHD-like symptoms in ASD [associated symptom or comorbidity, not the condition itself]	"Medications, including US Food and Drug Administration (FDA)-approved antipsychotics such as risperidone and aripiprazole, are used to manage comorbid irritability and aggression, stimulants and non-stimulants (e.g., methylphenidate, atomoxetine, clonidine, guanfacine) to treat ADHD-like symptoms, and melatonin for sleep disturbances." Recent Advances in Pharmacological Management of Autism: A Narrative Review
Atomoxetine	treating ADHD-like symptoms in ASD [associated symptom or comorbidity, not the condition itself]	"Medications, including US Food and Drug Administration (FDA)-approved antipsychotics such as risperidone and aripiprazole, are used to manage comorbid irritability and aggression, stimulants and non-stimulants (e.g., methylphenidate, atomoxetine, clonidine, guanfacine) to treat ADHD-like symptoms, and melatonin for sleep disturbances." Recent Advances in Pharmacological Management of Autism: A Narrative Review

Drug	Indication	Key evidence
Clonidine	treating ADHD-like symptoms in ASD [associated symptom or comorbidity, not the condition itself]	"Medications, including US Food and Drug Administration (FDA)-approved antipsychotics such as risperidone and aripiprazole, are used to manage comorbid irritability and aggression, stimulants and non-stimulants (e.g., methylphenidate, atomoxetine, clonidine, guanfacine) to treat ADHD-like symptoms, and melatonin for sleep disturbances." Recent Advances in Pharmacological Management of Autism: A Narrative Review
Guanfacine	treating ADHD-like symptoms in ASD [associated symptom or comorbidity, not the condition itself]	"Medications, including US Food and Drug Administration (FDA)-approved antipsychotics such as risperidone and aripiprazole, are used to manage comorbid irritability and aggression, stimulants and non-stimulants (e.g., methylphenidate, atomoxetine, clonidine, guanfacine) to treat ADHD-like symptoms, and melatonin for sleep disturbances." Recent Advances in Pharmacological Management of Autism: A Narrative Review

Recently approved or updated (FDA, EMA and CDSCO announcements)

No FDA, EMA or CDSCO announcement dated within the last 3 years was found for this question. This does not mean none exists.

2 further regulator page(s) were read but are not listed: the page states no date, or a date more than 3 years old, so they cannot be shown as recent.

Safety warnings (FDA boxed warnings)

Drug(s) (FDA label)	Boxed warning, in the label's words
Aripiprazole	"S: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS AND SUICIDAL THOUGHTS AND BEHAVIOURS WITH ANTIDEPRESSANT DRUGS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Aripiprazole is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."
Methylphenidate	"ABUSE, MISUSE, AND ADDICTION Methylphenidate hydrochloride extended-release has a high potential for abuse and misuse, which can lead to the development of a substance use disorder, including addiction."
Atomoxetine	"SUICIDAL IDEATION IN CHILDREN AND ADOLESCENTS Atomoxetine increased the risk of suicidal ideation in short-term studies in children or adolescents with Attention-Deficit/Hyperactivity Disorder (ADHD). Anyone considering the use of atomoxetine in a child or adolescent must balance this risk with the clinical need. Co-morbidities occurring with ADHD may be associated with an increase in the risk of suicidal ideation and/or behavior."
Risperidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Risperidone tablets are not approved for the treatment of patients with dementia-related psychosis."

These are the labels' own boxed warnings, not a complete safety review: read the full label. No boxed warning on the retrieved label of: Melatonin, Clonidine, Guanfacine.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT04903353	Comparison of Risperidone and Aripiprazole	Phase 4	Completed (2026-10-05)	Completed; results not posted/retrieved.
NCT07257939	NTI164; Placebo	Phase 3	Not yet recruiting (2026-09-30)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07285798	KarXT; KarX-EC	Phase 3	Not yet recruiting (2026-09-28)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06229210	Lumateperone	Phase 3	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT03279471	Sertraline; CBT/BIACA	Phase 2	Completed (2026-09-23)	Completed; results posted (outcome not retrieved here).
NCT07424404	KarXT; KarX-EC	Phase 3	Recruiting (2026-09-22)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04630847	Fecal Microbiota Transplant	Phase 1	Not yet recruiting (2026-09-18)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07439042	buspirone; Placebo	Phase 4	Recruiting (2026-09-17)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT03514784	BB-12 with LGG (Higher Dose); Placebo	Phase 1/2	Completed (2026-09-14)	Completed; results posted (outcome not retrieved here).
NCT03694275	Soticlestat	Phase 2	Completed (2026-09-10)	Completed; results posted (outcome not retrieved here). Specific syndrome record; do not generalise to the whole condition.
NCT05626959	NTI164	Phase 2/3	Completed (2026-09-10)	Completed; results not posted/retrieved.
NCT06368726	tDCS or Transcranial Direct Current Stimulation; Risperidone	Phase 1/2	Completed (2026-09-10)	Completed; results not posted/retrieved.

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Hume et al. (2021), Systematic Review:** Identified 28 evidence-based practices (EBPs) for ASD across 972 articles, recategorizing former EBPs and distinguishing manualized interventions [11].
2. **Sandbank et al. (2023), Meta-Analysis/Systematic Review:** Found significant RCT-supported effects of behavioral interventions on social communication (Hedges' $g = 0.58$) and challenging behaviors in early childhood ASD [5].
3. **Shi et al. (2026), Case Reports:** Bilateral dual-target deep brain stimulation in a 12-year-old with severe ASD led to sustained symptom reductions (ATEC/RBS-R improvements of 65.9–96.4%) over 12 months [1].

Conflicting evidence

Quan et al. (2022) reported *promising* but methodologically limited benefits of gluten-free/casein-free diets for ASD (reduced stereotypical behaviors, improved cognition), while **Anderson et al. (2023)** highlighted ethical concerns and negative experiences with applied behavior analysis (ABA), suggesting no consensus on dietary interventions' efficacy or acceptability.

What remains under investigation

- **Efficacy:** Head-to-head comparisons of behavioral interventions (e.g., ABA vs. naturalistic play) for long-term social communication outcomes.
- **Feasibility:** Scalability of deep brain stimulation (DBS) for severe ASD, given its experimental status and reliance on single-case studies.
- **Association:** Longitudinal microbiome-gut-brain axis studies to clarify probiotic efficacy beyond short-term diversity changes.
- **Effectiveness:** Real-world implementation of extended reality (XR) interventions (e.g., VR therapy) in diverse clinical settings.
- **Effectiveness:** Comparative effectiveness of music therapy vs. other non-pharmacological approaches (e.g., sensory integration) for specific ASD subgroups.

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- Excluded retracted publication: Efficacy of oral folinic acid supplementation in children with autism spectrum disorder: a (PMID 39243316, 2024). Retracted work is not used as evidence.
- 1 retrieved record(s) were excluded because the relevance reading judged them not to answer the question.
- The PubMed / Europe PMC search timed out, so it may be incomplete; the absence of evidence cannot be concluded from it.

Sources

1. Shi M, Hu J, Pei J, et al. Dual-target deep brain stimulation for severe pediatric autism: Sustained core symptom reduction in a 12-month case study. *Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology*. 2026 Dec. [PMID 42759099](#). DOI 10.1016/j.clinph.2026.2112403.
2. Hao D, Mansi H, Elkefi S, et al. Extended reality (XR) interventions in supporting children with autism: A systematic review. *International journal of medical informatics*. 2026 Dec. [PMID 42784985](#). DOI 10.1016/j.ijmedinf.2026.106719.
3. Erçelebi H, Özbudak P, Menderes D, et al. Altered sleep spindle characteristics in children with autism spectrum Disorder: A potential neurophysiological marker. *Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology*. 2026 Dec. [PMID 42753649](#). DOI 10.1016/j.clinph.2026.2112405.
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5. Sandbank M, Bottema-Beutel K, Crowley LaPoint S, et al. Autism intervention meta-analysis of early childhood studies (Project AIM): updated systematic review and secondary analysis. *BMJ (Clinical research ed.)*. 2023 Nov 14. [PMID 37963634](#). DOI 10.1136/bmj-2023-076733.
6. Siafis S, Çıray O, Wu H, et al. Pharmacological and dietary-supplement treatments for autism spectrum disorder: a systematic review and network meta-analysis. *Molecular autism*. 2022 Mar 4. [PMID 35246237](#). DOI 10.1186/s13229-022-00488-4.
7. Quan L, Xu X, Cui Y, et al. A systematic review and meta-analysis of the benefits of a gluten-free diet and/or casein-free diet for children with autism spectrum disorder. *Nutrition reviews*. 2022 Apr 8. [PMID 34617108](#). DOI 10.1093/nutrit/nuab073.
8. Geretsegger M, Fusar-Poli L, Elefant C, et al. Music therapy for autistic people. *The Cochrane database of systematic reviews*. 2022 May 9. [PMID 35532041](#). DOI 10.1002/14651858.CD004381.pub4.
9. Restoy D, Oriol-Escudé M, Alonzo-Castillo T, et al. Emotion regulation and emotion dysregulation in children and adolescents with Autism Spectrum Disorder: A meta-analysis of evaluation and intervention studies. *Clinical psychology review*. 2024 Apr. [PMID 38401510](#). DOI 10.1016/j.cpr.2024.102410.
10. Randell E, Wright M, Milosevic S, et al. Sensory integration therapy for children with autism and sensory processing difficulties: the SenITA RCT. *Health technology assessment (Winchester, England)*. 2022 Jun. [PMID 35766242](#). DOI 10.3310/TQGE0020.
11. Hume K, Steinbrenner JR, Odom SL, et al. Evidence-Based Practices for Children, Youth, and Young Adults with Autism: Third Generation Review. *Journal of autism and developmental disorders*. 2021 Nov. [PMID 33449225](#). DOI 10.1007/s10803-020-04844-2.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- [Autism Spectrum Disorder \(ASD\) Treatment](#)
- [Medication Treatment for Autism | NICHD - Eunice Kennedy Shriver National Institute of Child Health and Human Development](#)
- [Pharmacological treatments in autism spectrum disorder: a narrative review | Journal of Psychopathology](#)
- [Recent Advances in Pharmacological Management of Autism: A Narrative Review](#)
- [FDA drug label: Risperidone \(label effective 2026-06-25\)](#)

Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.